

Honors Intro Micro | Homework 4

Due Sunday Nov. 7

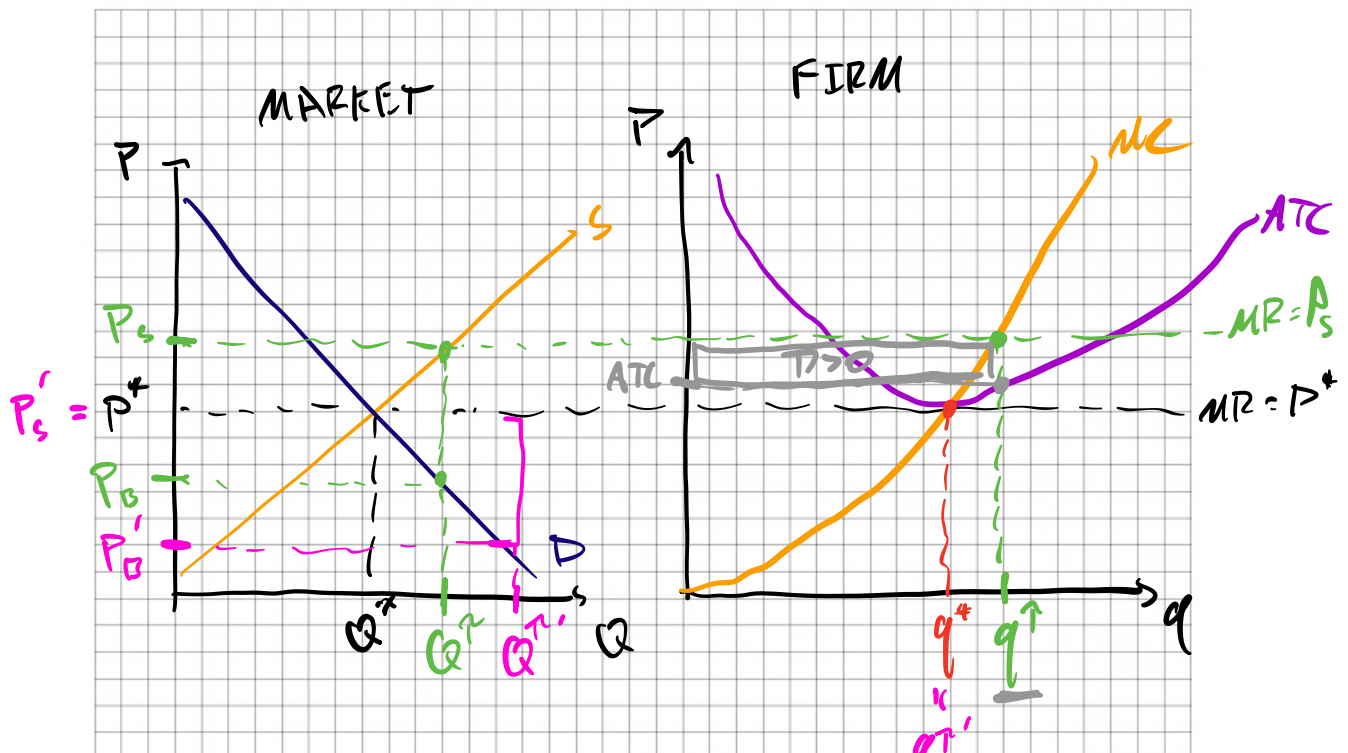
Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely; show your work so I can understand your thought process for partial credit; you are welcomed and encouraged to work in groups as long as your work is your own.

- Easy Entry
- Homogeneity

Question 1 | Subsidizing Bludgers

Quidditch is a game much beloved by the magical community. Since the rules are well established, the game equipment is standardized, and every equipment seller produces identical equipment. Bludgers are an important part of the game, are very easy to make, and require no specialized skills or expensive tools. Additionally, the number of sellers of Bludgers hasn't changed much in the over 500 years the sport has been played widely.

To spur interest in the sport, the Ministry imposed a subsidy on bludgers. Using a couple of graphs to illustrate your answer, explain what happened in the market because of this subsidy. Be sure to include all relevant components on the graph, and conclude in a long-run equilibrium.



1. $P^*, Q^*, q^*, \pi = 0$
2. $P_s, P_0, Q^*, q^*, \pi > 0$. This leads to firm entry.
3. Lastly, we conclude in long run equilibrium. $P'_s = P_0, Q^* > Q^* > Q^*$

and $q_{r'} = q^* < q^r$, $\pi = 0$.

Question 2 | Golden Snitches

The golden Snitch is another piece of equipment in the game of Quidditch. Since manufacturing the Snitch to the uniformity required for competition is incredibly difficult, everyone in the sporting world buys their golden Snitches from one seller: Ruffelbottom's Quidditch Supply. The demand for golden Snitches is given by:

$$P = 100 - Q$$

where prices are given in terms of galleons. Ruffelbottom's marginal cost of producing each Snitch is a constant 10 galleons, and there are no fixed costs. The marginal revenue per Snitch is:

$$MR = 100 - 2Q$$



Part A. Quantity

What is the profit maximizing number of Snitches Ruffelbottom should sell? Label on the graph above.

Part B. Price

What price should Ruffelbottom charge per Snitch? Label on the graph above.

Part C. Profit

What is Ruffelbottom's profit from selling Snitches? Label on the graph above.

Part D. Lumpsum Tax

The Ministry realized Ruffelbottom was capturing a great deal of welfare from the quidditch world, and decided to impose a lumpsum tax on the seller. Use a graph to show the affect this lumpsum tax had on the market.

